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Exercise 4

SQL Practice Exercise: Joins, UNION, Filtering, and Aggregates

1. SQL Joins

Question 1:

Retrieve all employees and their assigned projects using an INNER JOIN.

Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

Answer:

```
SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,
    p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM Employees e
INNER JOIN Projects p
ON e.EmployeeID = p.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

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Question 2:

Retrieve all employees and their assigned projects, including employees who have no projects using a **LEFT JOIN**.

Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

Answer:

```
SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,
    p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM Employees e
LEFT JOIN Projects p
```

ON e.EmployeeID = p.EmployeeID;

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

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Question 3:

Retrieve all projects and their assigned employees, including projects that have no employees using a **RIGHT JOIN**.

Return: ProjectID, ProjectName, Budget, Status, EmployeeID, FirstName, LastName, Department, Salary.

Answer:

SELECT p.ProjectID, p.ProjectName, p.Budget, p.Status,

e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary

FROM Employees e

RIGHT JOIN Projects p

ON e.EmployeeID = p.EmployeeID;

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity Audit	75000	Pending	1	John	Doe	IT	70000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	80000

Question 4:

Retrieve all employees and projects, including those without a match in either table using a **FULL OUTER JOIN**.
Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

Answer:

SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,

p.ProjectID, p.ProjectName, p.Budget, p.Status

FROM Employees e

FULL OUTER JOIN Projects p

ON e.EmployeeID = p.EmployeeID;

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

---

Question 5:

Retrieve a list of all unique cities where employees are located and project statuses.

Return: Location (Rename the column to Location using an alias).

Answer:

SELECT City AS Location FROM Employees

UNION

SELECT Status FROM Projects;

Location
New York
Los Angeles
Toronto
London
Sydney
Completed
Ongoing
Pending

Question6:

Retrieve a list of all cities where employees are located and project statuses, allowing duplicates. Return: Location (Rename the column to Location using an alias).

Answer:

SELECT City AS Location FROM Employees

UNION ALL

SELECT Status FROM Projects;

Location
New York
Los Angeles
Toronto
London
Sedney
New York
Completed
Ongoing
Pending
Ongoing
Completed
Pending

Question 7:

Retrieve employees who earn more than 70,000.

Return: EmployeeID, FirstName, LastName, Department, Salary.

Answer:

SELECT EmployeeID, FirstName, LastName, Department, Salary

FROM Employees

WHERE Salary > 70000;

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

Question 8:

Retrieve employees working in either IT or Finance departments.

Return: EmployeeID, FirstName, LastName, Department, Salary.

Answer:

SELECT EmployeeID, FirstName, LastName, Department, Salary

FROM Employees

WHERE Department IN ('IT','Finance');

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

Question 9:

Retrieve projects that are **not yet completed**.

Return: ProjectID, ProjectName, Budget, Status.

Answer:

SELECT ProjectID, ProjectName, Budget, Status

FROM Projects

WHERE Status != 'Completed';

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

Question 10:

Retrieve projects that have a **budget greater than 70,000** and are **not completed**.

Return: ProjectID, ProjectName, Budget, Status.

Answer:

SELECT ProjectID, ProjectName, Budget, Status

FROM Projects

WHERE Budget > 70000 AND Status != 'Completed';

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

Question 11:

Retrieve employees from New York OR Toronto, ordered by salary in descending order.

Return: EmployeeID, FirstName, LastName, Department, Salary, City.

Answer:

SELECT EmployeeID, FirstName, LastName, Department, Salary, City

FROM Employees

WHERE City IN ('New York','Toronto')

ORDER BY Salary DESC;

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

Question 12:

Retrieve the top 3 highest-paid employees.

Return: EmployeeID, FirstName, LastName, Department, Salary.

Answer:

SELECT EmployeeID, FirstName, LastName, Department, Salary

FROM Employees

ORDER BY Salary DESC

LIMIT 3;

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

Question 13:

Find the total salary per department, sorted in descending order.

Return: Department, TotalSalary (Rename SUM(Salary) as TotalSalary).

Answer:

SELECT Department, SUM(Salary) AS TotalSalary

FROM Employees

GROUP BY Department

ORDER BY TotalSalary DESC;

Department	TotalSalary

Finance	155000
IT	142000
Sales	65000
HR	60000

Question 14:

Find the **average salary per city**, but only include cities where the average salary is greater than **65,000**.

Return: City, AverageSalary (Rename AVG(Salary) as AverageSalary).

Answer:

SELECT City, AVG(Salary) AS AverageSalary

FROM Employees

GROUP BY City

HAVING AVG(Salary) > 65000;

City	AverageSalary
New York	75000
Toronto	75000

Question 15:

Count the **number of employees per department**, including only departments with more than 1 employee.

Return: Department, EmployeeCount (Rename COUNT(EmployeeID) as EmployeeCount).

Answer:

SELECT Department, COUNT(EmployeeID) AS EmployeeCount

FROM Employees

GROUP BY Department

HAVING COUNT(*) > 1;

Department	EmployeeCount
IT	2
Finance	2

Question 16:

Retrieve the **number of projects per status**, but only include statuses with **at least 2 projects**.

Return: Status, ProjectCount (Rename COUNT(ProjectID) as ProjectCount).

Answer:

```
SELECT Status, COUNT(ProjectID) AS ProjectCount

FROM Projects

GROUP BY Status

HAVING COUNT(*) >= 2;
```

Status	ProjectCount
Ongoing	2
Pending	2

Question 17:

Retrieve the **total project budget per employee**, but only for employees who are managing projects worth more than **150,000**.

Return: EmployeeID, FirstName, LastName, TotalProjectBudget (Rename SUM(Budget) as TotalProjectBudget).

Answer:

```
SELECT e.EmployeeID, e.FirstName, e.LastName,

    SUM(p.Budget) AS TotalProjectBudget

FROM Employees e

JOIN Projects p ON e.EmployeeID = p.EmployeeID

GROUP BY e.EmployeeID, e.FirstName, e.LastName

HAVING SUM(p.Budget) > 150000;
```

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000